

PH 701: Introduction to Molecular Simulations

Course Code	PH 701
Course Name	Introduction to Molecular Simulations
L-T-P-C	2-2-0-4
Prerequisites	Master-level courses in Physics/Chemistry
Intended for	Ph.D.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Classical statistical mechanics**
- **Ensembles**
- microcanonical, canonical, grand canonical ensembles ideal gas- harmonic oscillator – Spin Systems. Introduction to Stochastic process, Brownian Motion, Langevin equation, Fokker- Planck equation, Introduction to liquid state theory- pair distribution functions- structure factor- coherent and in-coherent scattering- Ornstein-Zernike correlation function Diffusion in a liquid- mean square displacement- self and collective van Hove correlation function – Intermediate scattering function and dynamics structure factor.
- **Programing in C and Fortran 95**
- Essential for programming in this course
- **Introduction of Monte Carlo methods**
- Value of using MC method, Gaussian distribution from 1d random walk, Metropolis algorithm for construction NVT ensemble, Implementation of ensemble using MC methods.
- **Project 1**
- Write a Monte Carlo simulation to simulate model liquid.
- **Introduction to Molecular dynamic simulations**
- Molecular dynamics simulations, Numerical integration of linear differential equations, Leap-Frog algorithm, Velocity Verlet algorithm, Periodic boundary condition one, two and three dimensions.
- **Project 2**
- Write a MD simulation code for simple liquids and for a polymer chain connected by harmonic spring.
- **Introduction to Brownian and Langevin dynamics simulations**
- Simple Brownian dynamics algorithm without hydrodynamic interactions. Langevin dynamics simulations.
- **Project 3**
- Write a Brownian dynamics code to simulate colloids in a solution and motion of single polymer chain.
- **Analysis data from simulations**
- Computation of radial distribution function, Structure factor, Time series analysis, Mean square displacement.
- **Project 4**
- Using trajectories produced from the earlier simulation to compute: Radial distribution functions. Mean square displacement of center of mass and monomers for a polymer chain. Computation of stress, stress correlation function and viscosity.

Text and Reference Books

1. R. K. Pathria, **Statistical Mechanics**
2. **Introduction stochastic process in physics and astronomy**, Rev. Mod. Phys., 1, 15, 1943.
3. J. A. Barker and D. Henderson, **What is liquid? Understanding the state of matter**, Rev. Mod. Phys. 587, 48, 1976.
4. J. P. Hansen and I. R. McDonald, **Theory of simple liquids**
5. D. A. McQuarrie, **Statistical Mechanics**.
6. M. P. Allen and D. J. Tildesley, **Computer simulation of liquids**
7. Daan Frenkel, **Understanding molecular simulation**.
8. D. C. Rapaport, **The art of molecular dynamics simulations**.
9. D. P. Landau and Kurt Binder, **A guide to Monte Carlo simulations in statistical Physics**.